

C++ <mutex> Header - Components Overview

Key Components in <condition_variable>

Component	Description
std::condition_variable	Used to block threads until a condition is met (with std::unique_lock).
std::condition_variable_any	More generic version that works with any BasicLockable type.
wait()	Blocks the current thread until notified and the predicate is true.
wait_for()	Blocks the thread for a specified time duration or until notified.
wait_until()	Blocks until a specified time point or notification.
notify_one()	Unblocks one waiting thread.
notify_all()	Unblocks all waiting threads.